

DARBALI NOUHAYLA

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📄 <https://github.com/DarbaliNouhayla/> 🏠 huggingface.co/darbalinohayla 🏆 [kaggle.com/shiranaihito](https://www.kaggle.com/shiranaihito)
Second-year Computer Science Engineering student at EHTP, specializing in the convergence of Applied AI (Computer Vision, RAG, Agentic Models) and production Software Engineering. Combining hands-on experience in model fine-tuning (YOLOv8), asynchronous architectures (FastAPI), and DevOps automation (Docker, Kubernetes), I am seeking a 2-month engineering internship for July – August 2026. Motivated to join an innovative team to deploy robust, high-performance, and scalable real-world solutions.

EDUCATION

Ecole Hassania des Travaux Publics (EHTP)

Master of Engineering in Computer Science

Sep. 2024 – May 2027

Casablanca, Morocco

CPGE - Preparatory Classes (MP)

Applied Mathematics, Computer Science, and Physics

Sep. 2022 – July 2024

CPGE Ibn Taher, Errachidia

High School Diploma (Baccalaureate)

Mathematical Sciences

2022

Honors (Highest Distinction)

Relevant Coursework

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|---------------------|------------------------|-----------------------|-------------------------|
| • Machine Learning | Analysis | • Algorithm Analysis | • Systems Programming |
| • Deep Learning | • Software Engineering | • Database Management | • Networking & Security |
| • Multivariate Data | • Data Structures | • Web Technologies | |

Professional Experience

Provincial Administration

July 2025 – August 2025

Full-Stack Development Intern — Provincial HRMS Project

- Architected an HR Management System using **Django** to digitize HR workflows for 100+ employees.
- Designed a **PDF report generation engine** (ReportLab) for exporting leave balance summaries.
- Implemented security via **RBAC** (Role-Based Access Control) to manage agent, validator, and admin privileges.
- Optimized **MySQL** database schemas to ensure the integrity of medical records and career history data.

Technical Projects

Traffic AI Morocco & ADAS Monitoring System | Python, YOLOv8, ByteTrack, OpenCV, OSRM

June 2026

- Built an automated smart city surveillance dashboard and pipeline using **YOLOv8** and **ByteTrack** for multi-object vehicle tracking.
- Fine-tuned a custom **YOLOv8** vision model to accurately detect and classify real-time driver behavioral anomalies, targeting drowsiness, sleepiness, and inattention.
- Developed a custom **Arabic OCR** pipeline for Moroccan license plate recognition and integrated an **OSRM** engine for live density-driven routing optimization.

ResearchPal AI: Agentic RAG Ecosystem | Python, LangChain, Unsloth, HuggingFace

March 2025

- Engineered a **RAG** assistant with semantic vector search (ChromaDB) for scientific corpora.
- Developed an **agentic loop (MCP)** for autonomous real-time synthesis of ArXiv research papers.
- Fine-tuned a **Llama 3.2-3B model (LoRA)** specialized on SciQ, optimizing VRAM usage by 60%.

Predictor Engine: Predictive Analytics | Python, Scikit-learn, hmmlearn, Pandas

March 2024

- Conducted a comparative study using **6 algorithms (SVM, Random Forest, KNN)** for football match prediction.
- Modeled dynamic performance via **Hidden Markov Models (HMM)** to capture latent team states.
- Performed feature engineering on 10 years of data with normalization pipelines using Pandas and Scikit-learn.

Audio Cloning & Deepfake Security (SecurAI) | Python, PyTorch, Beatrice v2, FastAPI

Feb. 2026 – Present

- Engineered a high-fidelity voice cloning pipeline using the **Beatrice v2** engine with targeted convolutions.
- Co-developed a multi-modal **deepfake detection** module targeting spatial, temporal, and acoustic anomalies.
- Simulated Autonomous Agentic Cyber-Attacks (**AACA**) via stream injection to test system resilience.

Technical Skills

Languages: Python, JavaScript, SQL, Java, HTML/CSS

AI / ML / Vision: PyTorch, YOLOv8, OpenCV, LangChain, Unsloth, ChromaDB, HuggingFace, ByteTrack, Scikit-learn

Web & Frameworks: FastAPI, Django, React, Three.js, GSAP, Bootstrap

Dev Tools & Engines: Docker, Kubernetes, OSRM, Git, GitHub, GCP, Vercel, VS Code